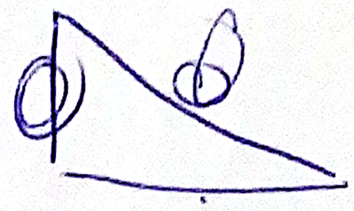
$$\sin^2(x) + \cos^2(x) = 1 \qquad / : \cos^2(x)$$



$$\tan^2(x) + 1 = \frac{1}{\cos^2(x)}$$

$$\int x^{-2} = -\frac{x^{-1}}{1} = -\frac{1}{x}$$

$$\int x^2 = \frac{x^3}{3}$$

$-\frac{e^{-2x}}{2} = -2\left(\frac{e^{-2x}}{2}\right) = e^{-2x} \quad -\frac{2}{3}+1-\frac{-2+3}{3} = \frac{1}{3}$

Problema 01

Calcular $\int \left(\frac{4}{3x^2} + 8e^{-2x} + \frac{2}{\sqrt[3]{x^2}}\right) dx$ $= \int \left(\frac{4}{3x^2} + 8e^{-2x} + \frac{2}{\sqrt[3]{x^2}}\right) dx$

$= \int \frac{4}{3x^2} dx + \int 8e^{-2x} dx + \int \frac{2}{x^{2/3}} dx = \frac{4}{3}\int x^{-2} dx + 2\int \frac{1}{x^{2/3}} dx + 8\int e^{-2x} dx$

$= -\frac{4}{3}\cdot\frac{1}{x} + 2\sqrt[3]{x} + 8\cdot\frac{-e^{-2x}}{2} = \frac{-4}{3x} + 2\sqrt[3]{x} - 4e^{-2x} + C$

.....

Problema 02

Calcular la integral $\int 3x^2\sqrt{5x^3-4}\,dx$

$\int 3x^2\sqrt{5x^3-4}\,dx =$

$u = 5x^3-4$
$du = 15x^2dx$
$\frac{du}{5} = 3x^2dx$

$= \frac{1}{5}\int \sqrt{u}\,du =$

$= \frac{1}{5}\cdot\frac{u^3}{3} = \frac{(5x^3-4)^3}{15} + C$

$uv - \int v\,du = \int \sin(u)\,du = -\cos(2x)$

$= \frac{1}{2}\int \frac{\sin(u)}{2}\,du$

$= 2\int \sin(2x)\,dx$

Problema 03

Calcular la integral $\int 4x \cdot \cos(2x)\,dx$

$$\int 4x \cdot \cos(2x)\,dx =$$

$u = 4x \rightarrow du = 4dx$

$dv = \cos(2x)dx \rightarrow v = \frac{\sin(2x)}{2}$

$u = 2x \quad du = 2dx$

$\cos(u)$

$\frac{1}{2}\int \cos(u)\,du = \frac{\sin(2x)}{2}$

$= \int 4x\cos(2x)dx = 2x\,\text{sen}(2x) - \int 2\sin(2x)\,dx$

$= 2x\,\text{sen}(2x) + \cos(2x) + C$

(B)

Problema 04

Resolver la integral $\int \text{sen}^2(x)\cos^3(x)\,dx$

$$\int \text{sen}^2(x)\cos^3(x)dx = \int \text{sen}^2(x)\cos^2(x)\cos(x)\,dx = \int \text{sen}^2(x)\left(1-\text{sen}^2(x)\right)\cos(x)\,dx$$

$u = \text{sen}(x)$

$du = \cos(x)dx$

$$= \int u^2(1-u^2)\,du = \int u^2 - u^4\,du = \int u^2 du - \int u^4 du$$

$$= \frac{u^3}{3} - \frac{u^5}{5} + C \Rightarrow \frac{\text{sen}^3(x)}{3} - \frac{\text{sen}^5(x)}{5} + C$$

(9)

$x^2+4x=-20 \Rightarrow /+4 \quad x^2+4x+4=-16$

$(x+2)^2+16=0$

Problema 05

Calcular la integral $\int \frac{-3}{x^2+4x+20}\,dx$ $= -3\int \frac{1}{(x+2)^2+16}dx = \frac{-3}{4}\arctan\left(\frac{x+2}{4}\right)+C$

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Problema 06

Aplicando fracciones parciales, calcular la integral: $\int \frac{2x-1}{(x+1)(x-3)}\,dx$

$\frac{2x+1}{(x+1)(x-3)} = \frac{A}{(x+1)} + \frac{B}{(x-3)} \quad / \cdot (x+1)(x-3) \Rightarrow 2x-1 = A(x-3)+B(x+1)$

Si $x=+3 \Rightarrow 5 = A(0) + B(4)$

$\frac{5}{4} = B$

Si $x=-1 \Rightarrow (2\cdot -1)-1 = A(-1-3)+B(0)$

$(-2)\cdot -1 = -4A$

$2 = -4A \Rightarrow A = -1/2$

X

$\int\left(\frac{\frac{-1}{2}}{x+1}+\frac{\frac{5}{4}}{x-3}\right)dx = \int\frac{\frac{-1}{2}}{x+1}dx + \int\frac{\frac{5}{4}}{x-3}dx$

$= \frac{5}{4}\int\frac{1}{x-3}dx - \frac{1}{2}\int\frac{1}{x+1}dx$

$= \frac{5}{4}\ln|x-3| - \frac{1}{2}\ln|x+1| + C$

4

8

Problema 07

Aplicado el método de sustitución trigonométrica, resuelva:

$$\int \frac{1}{\sqrt{x^2-9}}\,dx, \quad \text{Sabiendo que} \quad x = 3\sec(\theta)$$

$$\int \frac{1}{\sqrt{(3\sec(\theta))^2-9}}\,dx = \int \frac{1}{\sqrt{9\sec^2(\theta)-9}} 3\sec(\theta)\tan(\theta)\,d\theta =$$

$x = 3\sec(\theta)$

$dx = 3\sec(\theta)\tan(\theta)\,d\theta$

$$= \int \frac{1}{\sqrt{9(\sec^2(\theta)-1)}} 3\sec(\theta)\tan(\theta)\,d\theta = \int \frac{1}{\sqrt{\sec^2(\theta)-1}} \sec(\theta)\tan(\theta)\,d\theta$$

$$= \int \frac{1}{\sqrt{\tan^2(\theta)}} \sec(\theta)\tan(\theta)\,d\theta = \int \frac{1}{\tan(\theta)} \sec(\theta)\tan(\theta)\,d\theta = \int \sec(\theta)\,d\theta$$

(4)

$$= \ln|\sin(\theta)| + C \quad \therefore \quad = \ln\left|\frac{\sqrt{x^2-9}}{x}\right| + C$$